

4-side comparison — Qwen3 Coder MoE

Date: 2026-05-29

Sides

baseline — raw one-line goal + file content. No simplicio.
cli — the simplicio-cli 6-layer task contract (role/stack, goal, target, criteria, constraints, output shape).
cli + sp — same contract wrapped as user-input-X inside the simplicio-prompt v1.9 Tuple-Space + Yool runtime template.
cli + ag — same contract, with classified failure feedback (PHPUnit tail or missed regex patterns) fed back over up to 3 attempts. Mirrors simplicio task --verify.

Metrics

functional — real vendor/bin/phpunit --configuration phpunit.xml.dist on wesleysimplicio/sistema-sindico (PHP 8). Pass = full suite green.
regex — structural pattern match against the generated output (cheap proxy used by bench/run_offline.py).

Headline — pass rate per side, both metrics

Model	Metric	baseline	cli	cli+sp	cli+ag	Δcli	Δcli+sp	Δcli+ag
Qwen3-Coder-30B-A3B-Instruct	functional	33%	91%	91%	91%	+58	+58	+58
Qwen3-Coder-30B-A3B-Instruct	regex	36%	90%	98%	90%	+54	+62	+54
Qwen3-Coder-Next	functional	50%	83%	83%	91%	+33	+33	+41
Qwen3-Coder-Next	regex	44%	100%	94%	100%	+56	+50	+56

Agents verify-loop convergence

Lower attempts = the model resolved the case earlier; 1 means it passed on the first try. Max attempts capped per harness.

Model	Metric	avg attempts (cli+ag)
Qwen3-Coder-30B-A3B-Instruct	functional	1.17
Qwen3-Coder-30B-A3B-Instruct	regex	2.00
Qwen3-Coder-Next	functional	1.17
Qwen3-Coder-Next	regex	1.00

Cost & latency per call (functional bench)

Tokens/call averaged across the 12 cases. cli+ag burns more tokens AND wall-clock per case because it may run up to 3 attempts.

Model	Side	tokens/call	ms/call
Qwen3-Coder-30B-A3B-Instruct	baseline	1259	5030
Qwen3-Coder-30B-A3B-Instruct	cli	1449	4983
Qwen3-Coder-30B-A3B-Instruct	cli+sp	2412	4879
Qwen3-Coder-30B-A3B-Instruct	cli+ag	1579	5332
Qwen3-Coder-Next	baseline	1280	3571
Qwen3-Coder-Next	cli	1449	3485
Qwen3-Coder-Next	cli+sp	2416	3398

Qwen3-Coder-Next	cli+ag	1858	4210
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Per-task × model (functional, base / cli / cli+sp / cli+ag)

P = pass, . = fail. cli+ag suffix is the attempt count (1-3). 1 means no feedback loop needed; 3 with . means the loop ran to exhaustion and still failed (model-capability ceiling).

Task	Qwen3-C-30B-A3B-Instruct	Qwen3-C-Next
password_strength	./P/P/P(1)	./P/P/P(1)
password_require_symbol	./././.(3)	./P/./P(1)
env_get_int	./P/P/P(1)	P/P/P/P(1)
env_get_bool	./P/P/P(1)	././P/P(1)
admin_only_allowed_roles	P/P/P/P(1)	P/P/P/P(1)
rate_limit_bucket_key	./P/P/P(1)	./P/P/P(1)
base_repository_build_where_sql	./P/P/P(1)	./P/P/P(1)
router_has	./P/P/P(1)	P/P/P/P(1)
bugfix_password_policy_lowercase	P/P/P/P(1)	P/P/./P(1)
password_assess	P/P/P/P(1)	P/P/P/P(1)
base_repository_build_update_sql	./P/P/P(1)	././P/.(3)
router_extract_params	P/P/P/P(1)	P/P/P/P(1)

Fan-out N=200 (cli + sp subagent runtime)

The simplicio-prompt SubagentRuntime fires **200 parallel LLM calls per case** at temperature=0.7. Each output is scored by real PHPUnit AND structural regex. Modal vote = pass on the most-common output. The **regex - fn gap** column quantifies where the cheap regex proxy lies.

Model	Cases	fn per-att	rx per-att	fn modal	rx modal	gap	avg s/case
Qwen3-Coder-30B-A3B-Instruct	12	41%	92%	5/12	11/12	+51	43.3s
Qwen3-Coder-Next	12	92%	95%	12/12	12/12	+3	20.0s

Regex-vs-functional disagreement (gap >= 30 pts)

Each row below is a (task, model) where regex says PASS while PHPUnit says FAIL. This is the case-by-case evidence that the regex proxy is misleading for these particular failure modes.

Task	Model	rx	fn	gap
env_get_int	30B-A3B-Instruct	100%	0%	+100
env_get_bool	30B-A3B-Instruct	100%	0%	+100
admin_only_allowed_roles	30B-A3B-Instruct	100%	0%	+100
rate_limit_bucket_key	30B-A3B-Instruct	100%	0%	+100
base_repository_build_where_sql	30B-A3B-Instruct	100%	0%	+100
router_has	30B-A3B-Instruct	100%	0%	+100
base_repository_build_update_sql	Next	100%	61%	+39